MILITARY SPECIFICATION

LIGHT EMITTING DIODE ARRAY

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of common module, Light Emitting Diode Arrays, as follows:

Type I	SU-96/UA (180 element array)
Type II	SU-122/UA (120 element array)
Type III	SU-127/UA (90 element array)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-726	- Packaging Requirement Codes
MIL-STD-810	- Environmental Test Methods

DRAWINGS

USA ELECTRONICS COMMAND

DLSM-B-806559	- Light Emitting Diode Array, SU-96/UA
DL-5002475	- Light Emitting Diode Array, SU-127/UA
DL-5002478	- Light Emitting Diode Array, SU-122/UA

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Light Emitting Diode Array, herein referred to as the LED module, converts electrical signals into visible light within an infrared system.

3.2 Construction. The LED modules shall be constructed as specified herein and in accordance with the following:

Type I	DLSM-B-806559
Type II	DL-5002478
Type III	DL-5002475

3.2.1 Weight. Weight of the LED module shall not be greater than 0.35 pound.

3.3 First article. When specified in the contract (see 6.2) the contractor shall furnish first article units for inspection (see 4.3).

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings. Materials, parts, and processes not specified shall be selected by the contractor in accordance with MIL-P-11268.

3.5 Components. The LED module shall consist of an LED array of gallium arsenide phosphide (see 6.3.1 and 6.3.2) with internal normalization circuitry, connector frame, a field flattener lens, and an aperture.

3.5.1 LED array. Each LED array shall be operated (burned-in) for 150 hours prior to assembly in the LED module. Burn-in shall be performed by biasing the array so that minimum bias current is not less than 1.33 milliamperes. Ambient temperature need not be controlled during burn-in.

3.6 Performance characteristics. Unless otherwise specified herein, the LED module shall meet specified performance requirements at an ambient temperature of +23°C, ±2°C and all bias tolerances shall be ±0.1 Vdc.

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3.6.1 Radiated power output. The average element radiant power, defined as the sum of each element's radiant power divided by the number of elements, shall be as tabulated below:

Bias (Vdc)	Average radiant power x 10^{-6} (watts)		
	at -54°C	at +23°C	at +71°C
2.5	2.0	2.0	1.1
4.0	7.5	7.5, ±0.4	4.3
8.4	21.0	21.0	12.0

3.6.2 Input power. The total power supplied to the LED module with 4.0 Vdc applied (see 4.6.2) shall not be greater than 3.4 watts for type I, 2.3 watts for type II or 1.7 watts for type III.

3.6.3 Array configuration. The LED module shall consist of 180, 120 or 90 elements arranged in a configuration with pin-to-element connections as specified on the applicable drawing.

3.6.4 LED module tracking. The radiated power output of each element shall track the average element radiated power in accordance with the requirements as follows:

	Percent tracking		
Bias (Vdc)	@ -54°C	@ 23°C	@ 71°C
2.5	±15	±10	±15
4.0	±10	± 5	±10
8.4	±15	±10	±15

3.6.5 LED wavelength. The wavelength of light from each element shall be 660, ±10 nanometers with the LED module biased at 4.0 Vdc.

3.6.6 Emitting area. The emitting area dimensions shall be as specified on the applicable drawings. There shall be no more than 4 shape defects (see 6.3.3) in an LED array. The channels of the LED array as defined below shall be defect free of shape defects.

Type I - channels 31 through 150
Type II - channels 51 through 130
Type III - channels 71 through 110 and 141, 142 and 143

3.7 Environmental conditions.

3.7.1 Temperature shock. The LED module shall not be damaged (see 6.3.4) by sudden changes in temperature between -54°C and +95°C.

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3.7.2 High temperature. The LED module shall not be damaged under storage to +95°C, or continuous operation to +71°C, and shall operate for 30 minutes at +95°C.

3.7.3 Low temperature. The LED module shall not be damaged by operation to -54°C or storage to -62°C.

3.7.4 Shock. The LED module shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The LED module shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in figure 1.

3.8 Operational life (reliability). Over an accelerated operational life test of 500 hours when operated at a bias level of 4.0 Vdc and in a +50°C temperature environment, the average radiant power output of the diodes of the LED module shall not decrease by more than 20 percent. In addition, the LED module shall meet the tracking requirements of 3.6.4 after 500 hours of accelerated operational life testing. The 500 hour accelerated operational life demonstrates an MTBF of at least 6,804 hours.

3.9 Treatment. Unless otherwise specified, the LED module shall be cleaned and treated in accordance with the applicable drawing.

3.10 Nameplates and product marking. Unless otherwise specified, the LED module, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.11 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.9).

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4.3 First article inspection. When specified in the contract, the first article inspection shall be performed by the contractor on a minimum of 6 sample units.

4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Array burn-in	2/	3.5.1	4.3.1
Emitting area	2/	3.6.6	4.6.6
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Marking not as specified	2.5	3.10	MIL-M-13231
Workmanship not as specified	2.5	3.11	MIL-STD-454
Treatment not as specified	2.5 1/	3.9	
Dimensions not as specified	1.0	3.2	5002475 5002478 SM-D-806557
Diameter, interface (2.001" ref)			
Holes (4) thru cover, location			
Mark, alinement, location			
Pins, connector, location			
Field flattener lens			

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.
2/ Inspect every array in-process 100 percent.

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, perform the tests of table II in any order. Prior to start of tests specified in table II, LED modules shall be identified as to their applicable subgroups. Unless otherwise specified in the contract (see 6.2), three units shall be tested in subgroup 1 and three units in subgroup 2. Failure of any test on any unit shall constitute first article failure.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph	Subgroup 1	Subgroup 2
Radiated power output	3.6.1	4.6.1	X	X
Input power	3.6.2	4.6.2	X	X
Array configuration	3.6.3	4.6.3	X	X
LED module tracking	3.6.4	4.6.4	X	X
LED wavelength	3.6.5	4.6.5	X	X
Temperature shock	3.7.1	4.7.1	X	
High temperature	3.7.2	4.7.2	X	
Low temperature	3.7.3	4.7.3	X	
Shock	3.7.4	4.7.4	X	
Vibration	3.7.5	4.7.5	X	
Operational life (reliability)	3.8	4.8		X

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.g).

4.4 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the LED modules shall be operated at an ambient temperature of +23°C, ±2°C. External bias for performance tests shall be applied using a 205-ohm, ±1.0 percent resistor in series with each element.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted at ambient temperature only on all LED modules following successful inspection of 4.3.1. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that LED module.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Radiated power output	3.6.1	4.6.1
Input power	3.6.2	4.6.2
LED module tracking	3.6.4	4.6.4

4.5.2.2 Group B inspection. Not required.

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4.5.2.3 Group C inspection. Group C inspection shall be conducted on LED modules selected from LED modules which have passed the tests in 4.5.2.1. The sample(s) shall be tested in accordance with the inspections listed in table IV. Samples shall be selected in accordance with 4.5.2.3.1. Group C tests listed in table IV may be performed in any order.

4.5.2.3.1 Sampling for group C inspection. Group C inspection samples shall be selected as follows:

Monthly lot size	Number of sample units monthly
1-99	1
100-299	3
300-499	5
500-999	7
1000 or more	10

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Radiated power output	3.6.1	4.6.1
LED module tracking	3.6.4	4.6.4
LED wavelength	3.6.5	4.6.5

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2).

4.5.2.3.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to the tests of table III.

4.5.2.4 Group D inspection. Not required.

4.6 Test methods. Unless otherwise specified, all bias voltage tolerances shall be ±0.1 Vdc.

4.6.1 Radiated power output. The total radiated power output from each emitting element shall be measured at ambient, -54°C and +71°C with the LED module in its final assembled configuration. This test may be performed in conjunction with temperature tests 4.7.2 and 4.7.3. The emitted power is defined as the power radiated through the field flattener lens without applying any correction for the transmission losses due to the lens. The radiated power output shall be measured using a suitable detector of area (A) calibrated at 660 nanometers and positioned a distance (D) from the radiation source. The optical arrangement for this test is not critical, except that the solid angle of the cone of radiation subtended at the detector from the center element of the diode array shall not be greater than 0.2 steradian. The center of the detector shall be on the optical axis of the field flattener.

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The total radiated power output shall be calculated using the following equation:

$$P_T = P_A \left(\frac{\pi D^2}{A} + 1 \right)$$

P_T = Total radiated power (2π steradians)

P_A = Power measured at detector aperture

A = Sensitive area of detector

D = Distance of detector from radiation source.

Failure to meet requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Input power. The LED module forward bias voltage shall be adjusted to 4.0, ±0.1 Vdc and the current for each element shall be measured. The total power (P_T) shall be calculated using

$$P_T = \sum_{j=1}^{N} I_j (V - I_j R) \qquad \text{for } N = \begin{cases} 180 \text{ (type I)} \\ 120 \text{ (type II)} \\ 90 \text{ (type III)} \end{cases}$$

where I_j is the current for the jth element, V is the bias voltage (4.0 Vdc), and R is the series resistance (205 ohms, ±1 percent tolerance). All elements of the array need not be biased simultaneously. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Array configuration. Each element shall be separately energized at 4.0 Vdc bias and proper pin-to-element connections shall be verified. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 LED module tracking. The radiant power output for each element of the array shall be measured in accordance with 4.6.1 at forward bias voltage levels of 2.5, 4.0, and 8.4 Vdc. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 LED wavelength. The output wavelength of five randomly spaced elements shall be measured when biased at 4.0 Vdc. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Emitting area. The emitting area of all elements in the array shall be measured while the array is energized at 4.0 Vdc forward bias. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

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4.7 Environmental tests. The environmental tests shall be performed in accordance with MIL-STD-810 (see 6.2.e). Unless otherwise specified, the operating tests required by MIL-STD-810 before, during and after each environmental test shall be those specified in 4.6.1, 4.6.2, and 4.6.4.

4.7.1 Temperature shock. The LED module shall be tested in accordance with MIL-STD-810, Method 503, Procedure I except that the temperature of step 2 shall be -54°C and the temperature of step 1 and step 4 shall be +95°C. The LED module shall not be operated during temperature shock test (no measurements taken during this test). Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.7.2 High temperature. The LED module shall be tested in accordance with MIL-STD-810, Method 501, Procedure I except that the temperature of step 2 shall be +95°C and the LED module shall be operated during the final 30 minutes of step 3 at 4 Vdc bias (no measurements taken during step 3). The temperature of steps 4 and 5 shall be +71°C. The unit shall be operated at +71°C. Failure to meet the requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Low temperature. The LED module shall be tested in accordance with MIL-STD-810, Method 502, Procedure I except that the storage temperature of step 2 shall be -62°C and shall be maintained until temperature stabilization is reached. The low operating temperature of steps 4 and 5 shall be -54°C. The unit shall be operated at -54°C. Failure to meet requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Shock. The LED module shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure IV, Figure 516.1-1 (100 g's peak value at 11 milliseconds time duration) and Procedure V. The LED module shall not be operated during the shock test. Failure to meet requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Vibration. The LED module shall be tested in accordance with MIL-STD-810, Method 514.1, Procedures I and VIII except that the curve shown in figure 1 of this specification shall be used in lieu of the MIL-STD-810 curves. The LED module shall not be operated during vibration. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be one-sixth of the cycling time at each resonance. Failure to meet requirements of 3.7.5 shall constitute failure of this test.

4.8 Operational life (reliability).

4.8.1 First article operational life (reliability).

4.8.1.1 Test item - Type I units only.

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4.8.1.2 Test level.

Temperature +50°C, (+5 - 0°C)

Temperature cycling - none

Vibration - 2.2g, ±10 percent peak acceleration value at any non-resonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be normal to the plane of the LED.

Equipment ON-OFF cycling - Turn OFF 10 to 15 minutes during 3 hour cycle.

Duration - 500 hours on each sample.

4.8.1.3 Failure criteria. Ambient temperature tests 4.6.1, 4.6.2 and 4.6.4 shall be performed at least once prior to and after the test. Failure of any sample unit to meet the requirements of 3.6.1, 3.6.2 or 3.6.4 shall constitute failure of this test. Radiated power output requirement of 3.6.1 may be degraded by 20 percent maximum.

4.9 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-B-ØØ-00-NV-X-XX-Q-ED-A.

5.1.2 Level B. 2M-1-B-ØØ-00-NV-X-XX-Q-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The LED module is intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, date of this specification and type of LED required (see 1.1).
b. First article tests are required except for existing suppliers (see 6.3.5).

(1) Time frame for submission of first article test reports when first article tests are required.
(2) Time frame for approval of first article test reports when first article tests are required.

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c. Production delivery schedule.

(1) Defined in terms of monthly lots.
(2) Actions required relative to group C failures (see 4.5.2.3.2).

d. Level A or level B preservation and packaging (see section 5).
e. MIL-STD-810B shall be used for environmental tests of 4.7.
f. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
g. Deliver all first article samples to - Commander, USAERADCOM, ATTN: DELNV-CM, Fort Belvoir, VA 22060.
h. Group C inspection monthly sample size to be established based on cumulative totals of concurrent contracts.
i. Requirement for contractor-generated Government-approved acceptance test procedure.

6.3 Definitions.

6.3.1 Element. An element is defined as any one of the 180, 120 or 90 light emitting diodes of the LED module.

6.3.2 Array. An array is defined as a single line of 180, 120 or 90 elements arranged in the configuration shown on its respective drawing.

6.3.3 Shape defect. A shape defect of an individual element, measured when energized, is defined as an emitting area which varies by more than ±15 percent of the specified area or a length which varies by more than ±5 percent of the specified length as defined by the drawings (see 3.2).

6.3.4 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of LED module performance to values less than specified herein.

6.3.5 Existing suppliers. Suppliers who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12 month period. If a supplier cannot maintain the minimum monthly average, a first article test (less reliability) performed annually shall sustain the existing supplier's status when performed on 3 units manufactured by the supplier explicitly for this test. Upon implementation of this specification, suppliers who are then supplying equipment to specification B2-28A050103, Light Emitting Diode Array, with minimum monthly delivery rates specified above, shall be considered as existing suppliers.

CUSTODIAN:
Army - ER

PREPARING ACTIVITY:
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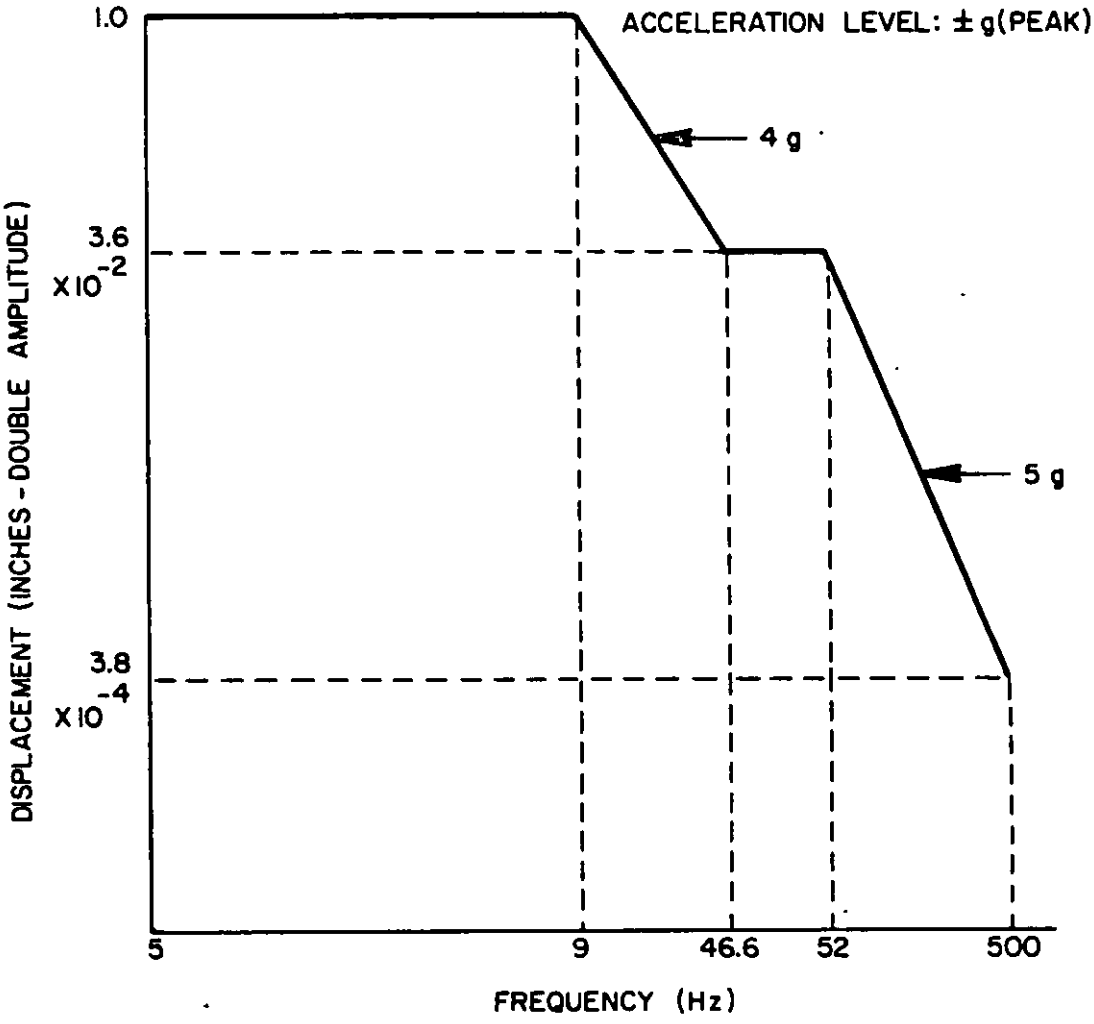


FIGURE 1. VIBRATION TEST PROFILE.

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DOCUMENT IDENTIFIER AND TITLE

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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